Project AHON: Executive Summary

Problem & Business Context

The Philippines is highly exposed to natural hazards, making disaster preparedness a critical priority for local government units (LGUs). Effective earthquake risk management requires an understanding of how seismic hazards intersect with population exposure, local government capacity, and available disaster risk reduction and management resources. However, these factors are often maintained in separate datasets, limiting decision-makers' ability to assess local conditions, identify preparedness gaps, and prioritize interventions.

Data & Approach

Project AHON addresses this challenge by integrating fragmented public-sector data into a unified, data-driven decision-support platform for earthquake preparedness. The project consolidates earthquake observations from the Philippine Institute of Volcanology and Seismology (PHIVOLCS), population data from the Philippine Statistics Authority (PSA), local government performance indicators from the Cities and Municipalities Competitiveness Index (CMCI), and disaster risk reduction funding information from the Local Disaster Risk Reduction and Management Fund (LDRRMF). The Philippine Standard Geographic Code (PSGC) serves as the common geographic reference for standardizing identifiers and linking information across datasets.

The data is then loaded and processed on Databricks in four steps:

1. Select and keep the original. Only relevant data is brought in, and an unaltered copy of every source is kept so any number can be traced back to where it came from.
2. Standardize. Location names are matched to PSGC so each city and municipality has exactly one official code, and spelling variants point to the same place.
3. Check. Data with serious quality problems is stopped, while data with smaller issues moves forward but is flagged. The result is a cleaned layer that is organized, validated, and matched to the right place.
4. Model. An LGU reference built from PSGC sits at the center, and every dataset connects to it while keeping its own details, so each LGU has one consistent profile.

The resulting analytical dashboard enables decision-makers to examine earthquake risk exposure, assess local preparedness, and identify areas requiring further attention, with population, disaster funding, fund utilization, and local government capacity indicators in a common view.

Key Insights

The dashboard facilitates comparisons across LGUs, allowing decision-makers to recognize differences in exposure and preparedness, identify capacity gaps, and examine practices from better-prepared localities that may be adapted to their own contexts.

Sta. Cruz, in Davao del Sur, was chosen as the focus purely because of what the data shows. Critical LGUs Risk exposure vs Budget Capacity was also determined wherein it is considered Critical when it has used less than one-fourth of its disaster fund while facing many earthquakes. Sta. Cruz is determined to

be High risk, and has low use of funds. Sta. Cruz recorded 354 earthquakes in the data, has about 105,000 residents, and has used only about 20% of its disaster fund, which makes it Critical. It was found out that its roads and infrastructure score is the weakest as well.

For comparison, T'boli, in South Cotabato, faces comparable earthquake risk with 355 recorded earthquakes, but has used 87% of its disaster fund, showing what a Critical LGU can move toward.

Recommendations

AHON is designed to complement, rather than replace, expert assessment and local validation. Its findings must be interpreted in consideration of data limitations and the specific circumstances of each community. Although the current scope focuses on earthquakes, we recommend that other hazards be explored and extend as well, as AHON's data integration approach provides a foundation for incorporating additional hazards in future development.

In addition to that, we recommend that findings be validated locally. AHON is designed to complement, rather than replace, the LGU's decision. Expert assessment and on-the-ground validation help explain what the data cannot, and results must be read in light of data limitations and each community's circumstances.

For LGUs, we would recommend that the dashboard be used to focus on places with high earthquake exposure and low fund utilization, such as Sta. Cruz. Aside from that, that they'll use the disaster funds they have already set aside, and close the capability gaps that keep funds from being used, starting with roads and infrastructure. Lastly, that they'll learn from better-prepared peers, such as T'boli, and adapt their practices to local conditions.

Ultimately, Project AHON aims to bridge the gap between fragmented data and local decision-making by making information on earthquake exposure, preparedness, and disaster funding more accessible and comparable. Through a shared analytical foundation, the platform supports LGUs in understanding their current conditions, identifying priorities for improvement, and learning from the preparedness of other communities, contributing to more informed planning and stronger local disaster resilience.